

How to Publish Videos to Social Media in Your AI Platform

This document explains the complete workflow for publishing AI-generated videos to social media platforms such as YouTube, Instagram, TikTok, and Facebook using official APIs, OAuth authentication, SEO generation, scheduling systems, and analytics tracking.

1. Create Developer Accounts

- Create Google Cloud project
 - Enable YouTube Data API v3
 - Create Meta Developer App
 - Enable Instagram Graph API
 - Configure OAuth credentials
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2. User Social Account Connection

- User clicks Connect YouTube or Instagram
 - Redirect to OAuth login screen
 - User grants permission
 - Receive access and refresh tokens
 - Store encrypted tokens securely
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3. Video Preparation

- Generate SEO title
 - Generate description
 - Generate hashtags and tags
 - Create AI thumbnail
 - Optimize video for platform
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4. YouTube Publishing Workflow

- Upload video via YouTube API
- Upload thumbnail
- Set privacy status
- Add playlist support
- Publish or schedule video

5. Instagram Publishing Workflow

- Create media container
 - Upload reel/video
 - Add captions and hashtags
 - Publish media
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6. Scheduled Posting System

- User selects publishing time
 - Store schedule in database
 - Celery worker processes task
 - Auto publish at selected time
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7. Analytics Tracking

- Track views
 - Track engagement
 - Track watch time
 - Track likes and shares
 - Display analytics dashboard
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8. Copyright Safety

- Check music copyright
 - Check duplicate videos
 - Analyze image similarity
 - Generate risk score
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9. Security Requirements

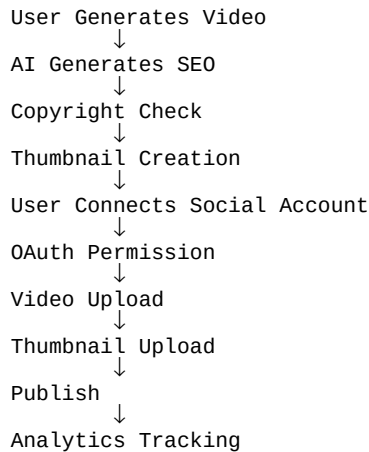
- OAuth 2.0 authentication
- JWT authentication
- HTTPS encryption

- • CSRF protection
 - • Rate limiting
 - • Secure token storage
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10. Recommended Technologies

- • Python Flask/FastAPI
 - • Celery and Redis
 - • AWS S3 storage
 - • YouTube API
 - • Instagram Graph API
 - • FFmpeg
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Complete Automated Publishing Workflow



Final Product Vision:

A complete AI-powered social media publishing system that automatically generates videos, optimizes SEO, checks copyright safety, securely connects user accounts, and publishes content across YouTube, Instagram, TikTok, and future platforms.